Samuel Santos

primary role

Principal Systems Engineer | Tech Lead SRE

technical core competencies

- Systems Architecture & Cloud: **Azure (Synapse, ADX), AWS (Lambda, S3, DynamoDB), Google Cloud Platform (GCP), Multi-Cloud Orchestration, Docker Containers.**
- Data Engineering & Telemetry: **Kusto Query Language (KQL), Azure Data Explorer (ADX), MS SQL Server, PostgreSQL, Cosmos DB, MongoDB, Entity Framework.**
- Methodologies & Specialties: **Site Reliability Engineering (SRE), FinOps & Cloud Cost Optimization, DevOps Automation, Low-Latency FinTech Systems.**
- API Architecture & Protocols: **Financial Information eXchange (FIX) Protocol, RESTful Web Services/APIs, Cross-Platform Interoperability, JSON/AJAX**
- Programming Languages: **C#, .NET (all versions), SQL, KQL, JavaScript.**
- Polyglot/Scripting Capabilities**: Bash, PowerShell, Python, C++ (Systems Interoperability), .NET CLI, AWS CLI, AWS Lambda Function, AWS S3 Bucket, Azure Functions, Azure Blob Storage.**
- Frontend Ecosystem: **JavaScript (ES6+), jQuery, TradingView Charts API, HTML5/CSS3.**

core skills

T-shaped person, Friendliness, Active Listening, Adaptability, Research, Self-taught Mindset, Driven for results, Hands-On, "go-to-person for hard riddles" (as my current leader likes to put it)

years of working experience

More Than 25 Years of Experience

current position

Working remote for Microsoft (SCHIE) as Tech Lead Product (SRE) since November 2024

education

B.E. Mechanical Engineering from Federal University of Rio de Janeiro - UFRJ
M.Tech. C.S. & System Project Management from Pontifical Catholic University - PUC-RIO

awards

Winning the IP2Location Programming Contest 2024
ABCM-EMBRAER Aerospace 2004 prize in undergraduate category

contact

samuel.santos.engineer@gmail.com

portfolio

https://samuel-santos-engineer.github.io/dev

social profiles

https://www.linkedin.com/in/samuel-santos-engineer
https://github.com/samuel-santos-engineer (My own labs)
https://github.com/v-safilho (Microsoft collaboration bits)

full name

Samuel Antonio Basto dos Santos Filho

location

Rio de Janeiro - RJ, Brazil, ZIP 22793-620

Bio

Seasoned Principal Systems Engineer and Technical Lead with over 25 years of experience delivering high-scale software architectures, resilient cloud infrastructure, and low-latency systems. **Currently driving core initiatives within Microsoft's Silicon, Cloud Hardware, and Infrastructure Engineering (SCHIE) division, collaborating with global teams across the US and India.** Recognized by leadership as the "go-to person for hard riddles" possessing a hands-on, investigative mindset to dissect legacy monoliths and resolve complex, systemic infrastructure issues. Expert at implementing automated FinOps frameworks, including cross-cloud garbage collector routines that eliminated over $20,000 in monthly infrastructure leaks.

Alumnus of UFRJ (Brazil's premier Tier-1 federal engineering university) and Robotics Lab researcher, honored with the prestigious national ABCM-EMBRAER Aerospace Prize. Holds post-graduate specialization in Computer Science from PUC-Rio (a globally top-ranked institution). Proven polyglot technical leader combining deep mastery of C# .NET, multi-cloud platforms (Azure Synapse/ADX, AWS, GCP), and enterprise databases with exceptional cross-cultural communication. **Adept at navigating international stakeholder alignment, influencing platform roadmaps, and driving immediate architectural ROI for high-stakes projects spanning Startups, FinTechs, and Global Tech Giants like Microsoft.**

Work Experience

Tech Lead Product and Site Reliability Engineering
Microsoft - SCHIE (*via Wipro Company*) | Microsoft HQ: Redmond, WA, USA (Remote)
November 2024 - Present
https://azure.microsoft.com/en-us/explore/global-infrastructure | *https://www.wipro.com*

Silicon, Cloud Hardware, and Infrastructure Engineering (SCHIE) division, certifying global Azure cloud data center readiness, hardware benchmarks, and cross-platform infrastructure orchestration.

Multi-Cloud FinOps & Cost Governance: Architected and deployed an automated multi-cloud garbage collector engine for AWS and GCP integration layers; successfully mitigated infrastructure provisioning leaks from orchestrator failures, saving over US$ 20,000 monthly in wasted compute resources.

Engineering Leadership & Hardware Certification: Serve as the technical lead guiding architectural decisions, engineering workflows, and cross-functional teams across Redmond and India (IDC) to orchestrate hardware certification routines for next-generation architectures (Intel Sapphire/Granite Rapids, AMD Genoa/Turin, ARM Cobalt 100/200).

Cross-Platform Orchestration Innovation: Designed a multi-cross-platform feature for the core Juno orchestrator, enabling global customers to execute client-server experiments across mixed silicon architectures (ARM64, X64) and operating systems (Linux, Windows).

High-Scale Telemetry & Observability: Engineer advanced queries and performance optimizations using Kusto Query Language (KQL) within Azure Data Explorer (ADX) to analyze, visualize, and debug complex infrastructure telemetry, application logs, hardware benchmarks, and system incidents. Contributed to optimize Kusto queries for benchmark analysis by architecting ETL pipelines within Azure Synapse Analytics for high-performance data transformation and consolidation.

High-Performance Workload Engineering: Unblocked a critical year-long Elasticsearch workload bottleneck within the Virtual Client platform by re-architecting an independent application installation flow; introduced a parallel-chunk downloading mechanism that reduced artifact transfer times from 90 minutes to under 2 minutes.

Automation & System Optimization: Authored Python-based storage automation scripts for Linux to implement dynamic RAID0 disk striping across attached cloud volumes, effectively unblocking high-throughput enterprise benchmark tools requiring massive continuous storage spaces.

Core Roadmap Influence: Resolved a critical CPU frequency ambiguity within ARM11 environments by engineering a dual-method continuous measurement framework; unblocked a high-priority, multi-team infrastructure program and directly informed the global hardware platform roadmap.

Operational Agility: Recognized by senior leadership for high-judgment execution and operational flexibility, successfully bridging the gap between deep software engineering and complex data center infrastructure operations under tight deadlines.

Selected Leadership Praise & Recognition:

“Samuel delivered a quick solution measuring the actual frequency with two methods – applicable to different environments. This work cleared ambiguity and informed the roadmap of a large program and many individuals.”
Alexander Frank, Engineering Leadership (May 2026)

“Samuel is consistently making good judgment calls, responding to inputs and opportunities with agility, and delivering impactful results timely. Samuel's work spans software development and complex operations tasks.”
Performance Appraisal Feedback (March 2026)

Principal Software Engineer & Founding Architect
PutCallBot Quant Trading System Fintech | Rio de Janeiro, Brazil (Remote)
September 2013 - November 2024
https://putcallbot.com

PutCallBot is a high-performance quantitative trading software solution that evaluates real-time Bovespa exchange market data to execute automated day-trading strategies, scaling from a private portfolio tool into a certified web platform serving high-net-worth clients.

Greenfield Product Engineering: Engineered a high-performance quantitative trading software solution entirely from scratch as a greenfield project; scaled the platform from a private portfolio tool into a certified web platform serving high-net-worth clients and processing multi-million BRL volumes.

High-Scale Financial Data Engineering: Eliminated critical I/O bottlenecks by replacing legacy relational database layers with a custom, ultra-fast object-oriented flat-file storage system; optimized data layers to seamlessly process a 200 GB real-time ticker database expanding at 40 MB per market day.

Low-Latency Broker Integrations & Compliance Workarounds: Spearheaded platform integrations with IT engineering teams at Brazil's largest brokerages (BTG Pactual, Genial, XP Group) using RESTful APIs and the Financial Information eXchange (FIX) protocol; successfully saved business continuity when XP Group suspended direct FIX/VPN endpoints for non-institutional software vendors.

Proprietary Algorithmic Order Routing Layer: Designed and developed a custom, language-agnostic order routing architecture capable of handling multi-account broker sync, manual order overrides, and virtual simulated order environments; engineered a proprietary workaround natively inside the MetaTrader platform using C++ as a tactical low-level tool to re-establish real-time broker communication, successfully bypassing the endpoint shutdown and safeguarding the platform's survival.

Quantitative Collaboration & Backtesting: Partnered with quantitative PhD researchers to implement algorithmic trading logic; built an independent, isolated Backtest Module allowing users to safely debug, optimize, and simulate trading robots against massive historical market data sets.

Cloud Cost Optimization & Orchestration: Designed the entire AWS cloud topography and engineered a specialized resource-scheduling engine utilizing the AWS .NET API; dynamically orchestrated low-cost spot server instances on-demand for heavy backtesting workloads, significantly reducing operational cloud overhead.

Full-Stack Dashboard Ecosystem: Developed a secure web portal and custom JavaScript framework managing real-time JSON/AJAX data flows, multi-broker configurations, and interactive data visualization utilizing the TradingView charts library.

Predictive Modeling R&D: Pioneered an R&D initiative introducing Deep Learning and Machine Learning techniques into the platform's prediction models; engineered training pipelines utilizing ML.NET, Python (Pandas, Jupyter), Keras, and TensorFlow to enhance predictive analytics accuracy.

Brownfield Legacy Modernization: Executed a comprehensive brownfield modernization initiative as the final key contribution to the project, successfully migrating and refactoring the legacy monolithic trading ecosystem into a highly resilient, service-oriented AWS cloud and .NET microservices architecture.

Principal Full Stack Engineer, C# .NET JavaScript, BoxBrazil, Logistics Startup
October 2010 - September 2013. Hybrid work from Rio de Janeiro, Brazil

BoxBrazil was an international e-commerce and cross-border logistics startup handling parcel forwarding and customs clearance from the US to the Brazilian market, scaling to 20,000+ registered customers.

Team Leadership & Engineering: Recruited, built, and led the initial IT engineering team from scratch, establishing development workflows to build the core e-commerce, back-office CRM, and automated customer service ticketing platforms.

Financial & Regulatory Engineering: Designed and built a mission-critical client-server tax engine that calculated complex customs fees, automated Brazilian IRS tax forms, and generated standardized banking barcode payment slips for automated processing.

API Architecture & Integration: Engineered robust RESTful Web Services and data pipelines integrating the platform with global and local providers, including Amazon Marketplace (MWS), DHL, Correios, PayPal, AWS SES, and major Brazilian banking APIs.

Security & Compliance Governance: Spearheaded the implementation of strict PCI-DSS compliance data security requirements for credit card transactions, deploying vulnerability assessments and penetration testing frameworks.

Full-Stack Stack Ownership: Owned the technical design and end-to-end development using C#, .NET Framework, SQL Server, Entity Framework, AWS Cloud Services, and JavaScript (jQuery/AJAX).

Staff Software Engineer, Itau Previtec (Pension Fund Management)
July 2000 - October 2010. Rio de Janeiro, Brazil

Itau Previtec (later acquired by Sinqia) was a market-leading enterprise software provider formed by a joint *venture with Itaú Bank, the largest financial institution in Brazil, serving over 100 enterprise clients including Fortune 500 giants like GM, Ford, and Nestlé.*

Engineering Leadership: Promoted to Staff Engineer and Project Manager, leading a cross-functional development team to scale and maintain core pension fund management systems serving major enterprise clients like GM, Volkswagen, Ford, and Nestlé.

Enterprise Architecture: Designed and deployed a custom JavaScript framework and corporate data management infrastructure using Business Intelligence (BI) and Microsoft OLAP Data Warehousing to empower executive-level decision-making.

Mobile Innovation: Conceptualized, built, and launched the company's very first mobile product using WAP protocols, creating a high-margin, profitable product line that allowed directors to access real-time market valuations on the go.

Technical Due Diligence: Served as the core technical point of contact during joint-venture M&A audits between SFR Previtec and Itaú Bank.

Education

- Computer Science & System Project Management @ Pontifical Catholic University of Rio de Janeiro - PUC
  Master of Technology - M.Tech - postgraduate degree
  2005-2006, Rio de Janeiro, RJ - Brazil
- Mechanical Engineer @ Federal University of Rio de Janeiro - UFRJ
  Bachelor of Engineering - B.Eng - undergraduate academic degree
  1995-2001, Rio de Janeiro, RJ - Brazil

Award

- Winning the IP2Location Programming Contest 2024
  November 2024, IP2Location Company, https://contest.ip2location.com/winners
  AWS Lambda Serverless Function to control IP requests and web resources using Geolocation by Country and City. https://github.com/samuel-santos-engineer/IPGeoGuard
- ABCM-EMBRAER Aerospace 2004 prize in undergraduate category
  2001-2003, UFRJ Robotics Lab (LabRob), Rio de Janeiro, RJ - Brazil
  A Digital System for Measurements in Gypsum Molds for Orthodontics Mechanical Engineering Department

Financial Market & Quantitative Certifications

- Capital Markets & Securities Analyst For Trading Floor Certification
  1998-1999, Rio de Janeiro Stock Exchange, Rio de Janeiro, RJ - Brazil
  Economics for Capital Markets; Financial and Statistical Calculation; Asset Classes; Financial Instruments and Markets; Equity Markets Trend Analysis; Portfolio Management; Brazilian Securities Law; Structure and Dynamics of a Trading Floor Negotiation.
- Financial Mathematics with HP 12c
  1997, 40 hours, Rio de Janeiro Stock Exchange, Rio de Janeiro, RJ - Brazil
  Financial Fundamentals, Simple interest, Compound interest and Amortization, Discounted Cash Flow Analysis, Bond and Depreciation Calculations.